

BRAINPOWER EDUCATION

Specialist Mathematics Units 1/2 - Further Trig 2026

Assessment Guide / Marking Scheme - 33 marks

Q1a - 1

$\cos x = -2/3$ and x is in quadrant III, so $\sin x = -\sqrt{5}/3$ and $\tan x = \sqrt{5}/2$.

Mark	Evidence required
1	$\tan x = \sqrt{5}/2$.

Q1b - 1

$\tan(2x) = 2\tan x / (1 - \tan^2 x) = -4\sqrt{5}$.

Mark	Evidence required
1	$-4\sqrt{5}$.

Q1c - 2

$1/(1 - \sin x) + 1/(1 + \sin x) = 2/(1 - \sin^2 x) = 2\sec^2 x$.

Mark	Evidence required
1	Combines fractions correctly.
1	Uses Pythagorean identity.

Q2a - 3

$h = (\sqrt{2} + 1)\sin x - \cos x$. Thus $R = \sqrt{4 + 2\sqrt{2}}$ and $\tan(\alpha) = 1/(\sqrt{2} + 1) = \sqrt{2} - 1$.

Mark	Evidence required
1	Correct coefficient comparison.
1	Correct R .
1	Correct $\tan(\alpha)$.

Q2b - 3

Correct sketches on $[0, \pi]$, intersecting at $x = 5\pi/8$; $y = \sqrt{(2 + \sqrt{2})}/2$.

Mark	Evidence required
1	Correct f sketch.
1	Correct g sketch.
1	Correct labelled intersection.

Q2c - 1

$x = 5\pi/8$.

Mark	Evidence required
1	Correct exact x -coordinate.

Q3a - 2

Domain $[0, 1]$; range $[0, 2\pi]$.

Mark	Evidence required
1	Domain $[0,1]$.
1	Range $[0,2\pi]$.

Q3b - 2

If $\theta = \arcsin(-3/5)$, $\tan \theta = -3/4$. Hence $\tan(2\theta) = -24/7$.

Mark	Evidence required
1	Correct triangle/ $\tan(\theta)$ and double-angle setup.
1	$-24/7$.

Q3c - 2

$x = (1 + \sqrt{5})/4$. Direct substitution gives $4x^2 - 2x - 1 = 0$.

Mark	Evidence required
1	Correct substitution.
1	Simplifies to zero.

Q4a - 3

$\sin 3t / \sin t = 3 - 4\sin^2 t$ and $\cos 3t / \cos t = 4\cos^2 t - 3$. Difference $= 6 - 4(\sin^2 t + \cos^2 t) = 2$.

Mark	Evidence required
1	Correct triple-angle identities.
1	Correct quotient simplification.
1	Obtains 2.

Q4b - 3

$\cos 3t = \cos 2t$ gives $3t = 2k\pi \pm 2t$. In $(0, \pi]$, $t = 2\pi/5$ or $4\pi/5$.

Mark	Evidence required
1	General cosine equation.
1	Solves branches.
1	Both domain-valid solutions.

Q5a - 2

$\tan(x/2) + \cot(x/2) = 1/[\sin(x/2)\cos(x/2)] = 2\csc x$.

Mark	Evidence required
1	Combines terms.
1	Uses $\sin x = 2\sin(x/2)\cos(x/2)$.

Q5b - 3

$2\csc x = 4 \rightarrow \sin x = 1/2 \rightarrow x = \pi/6$ or $5\pi/6$.

Mark	Evidence required
1	Reduces to $\sin x = 1/2$.
1	Finds candidates.

Mark	Evidence required
1	Both exact solutions.

Q6a - 3

Using double-angle identities, the ratio becomes $2\cos x(\cos x + \sin x) / [2\sin x(\sin x + \cos x)] = \cot x$ where defined.

Mark	Evidence required
1	Correct double-angle substitutions.
1	Correct factorisation.
1	Obtains $\cot x$.

Q6b - 2

$\cot x = 3 \rightarrow \tan x = 1/3$. Thus $x = \arctan(1/3)$ or $\pi + \arctan(1/3)$.

Mark	Evidence required
1	$\tan x = 1/3$.
1	Both solutions in $(0, 2\pi)$.